



Figure 5: Stuxnet makes use of vulnerabilities to reach the target PLCs

space. Then it checks for the appropriate platform (Export 15 Stuxnet routine to check for appropriate Windows version) to execute the code; if the machine is already infected, zero day vulnerabilities are used for escalating privileges. Finally, it installs (Export 16 Stuxnet routine for installation) two kernel drivers (*Mrxnet.sys* and *Mrxcls.sys*) - one for execution of Stuxnet after reboot and the other to hide the files.

Kernel module: Kernel module consists of two dropped driver files, namely, *Mrxnet.sys* and *Mrxcls.sys*. The latter is a driver used to mark itself as startup in the early stages of a boot. This driver takes responsibility for reading the registry and contains information for injecting Stuxnet code. *Mrxnet.sys* is used to hide the Stuxnet files; it creates the device object and attaches it to the system object (*s7otbxdc.dll*) of step7 to monitor all the requests.

Interestingly, as in the case of sabotaging the Iranian nuclear programme, Stuxnet monitored and recorded the normal operating frequency of centrifuges and fed the recorded data to the WinCC monitor during the attack. If the operator found the centrifuges are not operating normally, then Stuxnet shut itself down to a safe mode to remain undetected.

How to figure out if you are a victim of APT attacks

Here's a list of things you must do to spot an APT attack before it is too late:

- An APT may go unnoticed by a single antivirus/intrusion detection system; multi-level defence is required.
- Analysing network layered packets is a good way to keep a check.
- Log analysis from various sources with co-relation may help in early detection.
- You must monitor end points for suspicious behaviour.
- Good asset management should be in place to guard critical systems.
- Monitoring critical assets is very important. **END** 

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By: Sibi Chakkaravarthy Sethuraman

The author holds an M. Tech degree in computer science and engineering, and is currently pursuing a PhD at Anna University. Currently he is with AU-KBC Research Centre, MIT, Chennai. His research interests include network security, wireless security, wireless sensor networks, cloud security, etc. He can be reached at sb(DOT)sibi(AT)gmail(DOT)com.

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